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An Efficient Cluster Based Routing in Wireless Sensor Networks Using Multiobjective-Perturbed Learning and Mutation Strategy Based Artificial Rabbits Optimisation

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ABSTRACT

Wireless sensor networks (WSNs) is a wireless system including the set of distributed sensor nodes used for physical or environmental observation. A network energy expenditure is considered as a significant concern because of battery restricted sensors of the WSN. Clustering and multi hop routing are considered as effective approaches to enhance the network lifecycle and communication. Achieving the anticipated objective of reducing the energy expenditure, thereby increasing the network lifecycle, is considered as an optimisation issue. In recent times, a nature inspired meta-heuristic approaches are extensively utilised for solving the different optimisation issues. In this context, this research aims to accomplish the objective by proposing the multiobjective-perturbed learning and mutation strategy based artificial rabbits optimisation namely M-PMARO for an optimum cluster head (CH) selection and route discovery. The proposed M-PMARO incorporates an experience based perturbed learning (EPL) and mutation strategy to identify the capable regions over the search space for enhancing the exploration and avoiding the local optima issue. To formulate the multiobjective, the residual energy, average intracluster distance, average base station (BS) distance, CH balancing factor (CHBF) and node centrality are incorporated for optimum CH discovery while the residual energy and average BS distance are considered for multi hop routing. The M-PMARO is analysed based on alive nodes, dead nodes, energy expenditure, throughput and data received in BS and network lifecycle. The viability of M-PMARO is validated by comparing it with existing approaches such as fitness based glowworm swarm with fruitfly algorithm (FGF), energy balanced particle swarm optimisation (EBPSO), improved bat optimisation algorithm (IBOA), graph neural network (GNN) and fuzzy logic and particle swarm optimisation (PSO) based clustering routing protocol namely PFCRE. The alive node count of M-PMARO is 100 for 1200 rounds, which is higher than the EBPSO.

1 | Introduction

Wireless sensor network (WSN) is a distributed self-organising network which has numerous small, inexpensive and low power sensors that are located in harsh environments for monitoring

purposes [1–3]. WSN is one of the developing technologies of wireless ad hoc networks utilised in daily lives, which gathers, processes and communicates the information obtained from the physical environment. Hence, the WSN links the physical world with the digital world with cost effective and rapid

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communication [4]. Sensors of WSN sense, observe, track and identify the physical environments. Sensors are effective enough for ease of information broadcasting over wireless links. The gathered information is broadcasted to the base station (BS) of the data sink [5]. The sensors are incorporated with the microprocessor and electronic gadgets to link with the remaining devices, whereas the battery-powered energy supply is utilised in the sensor [6]. WSN discovered its applicability in various real-time applications, because of its low-cost, scalable and compact feature of sensors [7]. Some of the applications are industrial process control, environmental monitoring, domestic monitoring, military and lately in the internet of things [8].

Sensors have restricted battery capacity; therefore the lifecycle of the network is defined by the energy expenditure of installed sensors. A recharging or replacing of the sensor's batteries is not possible once it is deployed on the network [9, 10]. Hence, an important goal of WSN is to minimise the energy expenditure of sensors [11]. The clustering approach is developed with the goal of enhancing the energy efficiency, improving communication and increasing lifecycle [12–14]. In clustering, an entire network is separated into small regions, namely clusters, where a cluster head (CH) is chosen from each cluster. The chosen CHs collect the information from the sensors of the local cluster and it is broadcasted to the BS [15, 16]. After CH collected and aggregated the data from the sensor of the local cluster, the CH broadcasts the information to BS through one-hop or multi-hop transmission, which is represented as inter-cluster communication [17]. The fundamental idea of the clustering routing approach is concentrated on choosing the CH to obtain the energy load balancing between the sensors of the network [18].

While previous research has designed the optimisation approaches in clustering and routing, there is a gap in the research regarding the cluster balancing and failure to ensure load balancing in WSN. This gap motivates this research to design the M-PMARO as a promising solution to address the aforementioned issues.

1.1 | Contributions

The proposed Multiobjective-Perturbed Learning and Mutation Strategy based Artificial Rabbits Optimization (M-PMARO's) contributions are as follows:

- The combination of experience based perturbed learning (EPL) and mutation strategy used in artificial rabbits optimisation (ARO), transforms it into M-PMARO that helps to obtain the optimum solution, i.e., CHs and routing path. The EPL effectively combines the exploration and exploitation, which helps in enhanced searching for the optimum solution while the mutation strategy is used to avoid the local optima issue.
- The multiple objective functions, such as residual energy, average intracluster distance, average BS distance, CHBF and node centrality are used in M-PMARO for optimum CH selection, which leads to minimise the energy and achieve the reliable communication.

- Further, the multi hop path selection is obtained using M-PMARO according to the residual energy and average BS distance.

This is how the rest of the paper is organised. Section 2 deliberates the related work about the clustering and routing in WSN. The preliminaries are given in Section 3. The proposed M-PMARO is described in Section 4 and an assessment of proposed M-PMARO is given in Section 5. Finally, the Section 6 concluded the research.

2 | Related Work

Many WSN researches are concentrated on performing an effective CH and path discovery for reliable communication with less energy expenditure. Some recent research related to the swarm intelligence and fuzzy based methods is given below.

2.1 | Swarm Intelligence Based Methods

Elshrkawey et al. [19] presented the routing algorithm (RA) based on reposition particle swarm optimisation (RPSO) for enhancing the lifecycle and throughput. The developed RPSO was used to preserve the particles that faced the issue of local minima. The RA-RPSO was considered the sensor density, remaining energy, space among CH and BS, and history of CH to select the CHs from the sensors. The cluster balancing was mandatory to additionally minimising the energy expenditure.

Barnwal et al. [20] presented the metaheuristics cluster-based routing technique (MHCRT). This MHCRT was used as the whale moth flame optimisation to accomplish the clustering according to the balancing factor, intra cluster distance, inter cluster distance and energy. Further, a path identification using improved African buffalo optimisation was done based on the distance and energy. The MHCRT was enhanced the energy efficiency based on appropriate clustering and routing. However, a centrality of nodes was essential for achieving reliable communication over the network.

Chaurasia et al. [21] presented a cluster-based routing using an energy efficient meta heuristic approach in WSN. This work used the dragonfly algorithm (DA) to choose the optimum CH and path. The exploitation of DA was utilised to choose the CH according to the energy, distance and node density. Next, the random walk/ levy distribution function was used to select the path according to the exploration. This DA based clustering was required to consider the centrality of nodes for achieving reliable communication.

Mishra and Yadav [22] used the butterfly optimisation algorithm (BOA), particle swarm optimisation (PSO), and ant colony optimisation (ACO) for minimising the power usage and enhance the lifespan. At first, the CHs were chosen by using the node's remaining power, centrality of node, degree of node and interspace among the nodes and the base station in BOA. Nest, PSO formed the clusters by using the interspace among CH and BS. Finally, the ACO was used to identify the path. But, the balancing between the clusters was mandatory to improve the network lifespan in the next level.

Dattatraya and Rao et al. [23] developed the fitness based glowworm swarm with fruitfly algorithm (FGF) for performing an effective CH selection. The FGF was an integration of glowworm swarm optimisation and fruitfly optimisation approaches. An optimum CHs from the network were chosen by considering the energy, delay and distance in the FGF. This work was mainly concentrated only on CH discovery and it failed to concentrate on routing in the network.

Xiuwu et al. [24] developed energy balanced PSO (EBPSO) for minimising the energy expenditure, balancing the load and increasing the lifespan of the WSN. The developed EBPSO was restricted to the potential participating nodes in the CH discovery, as well as the nodes with lesser than average energy weren't allowed to contribute in the CH discovery. Next, each node's weight coefficient was computed during cluster generation. Further, the multi hop routing was developed based on the energy and distance. The EBPSO considered only the distance and energy while choosing the CH. An inappropriate selection of fitness measures affected the data broadcasting in the WSN.

Janakiraman [25] presented the improved bat optimization algorithm (IBOA) for performing the CH discovery to reduce the energy expenditure. Further, an enhanced artificial bee colony based sink mobility approach is developed for enhancing the data broadcasting over the network. However, the energy balancing among the clusters was not considered while performing the CH selection.

Daneshvar and Mazinani [26] developed the modified salp swarm algorithm (MSSA) to perform clustering and routing according to the multidimensional knapsack problem. The SSA was used to choose the relays and CHs were first node die (FND) were considered as a fitness function. Next, an adaptive routing was used to choose an appropriate amount of relays for each CH. Every relay belonged to one CH, hence there was a distinct group of chains between BS and any CH. This helped to balance the energy expenditure between the nodes. The load balancing among the clusters was mandatory for an additional decrement in energy expenditure.

2.2 | Fuzzy and Artificial Intelligence Related Methods

Ambareesh et al. [27] presented an optimum routing through CH and BS using the fuzzy technique for preference by similarity to ideal solution (Fuzzy-TOPSIS) based hybrid jarratt butterfly optimisation (FTOPSIS-HJBO). In this work, the FTOPSIS was integrated with the HJBO for updating their location to identify the optimum solution, which was optimum routing over the network. The developed FTOPSIS-HJBO considered the traffic, energy and distance factors to determine the optimum route. Therefore, the data was effectively broadcasted by selecting an optimum CH and path. The energy expenditure of the nodes needs to be minimised for further increasing the lifecycle of the network.

Rawat et al. [28] developed the fuzzy logic model and PSO for enhancing the lifespan of WSN. Here, the fuzzy logic was used to choose the CHs according to neighbour node count, interspace

to neighbours and energy ratio. Further, the PSO considered the distance and energy factor to perform the optimum node discovery for routing purpose. The inappropriate fitness function of clustering and routing affected the packet broadcasting and energy expenditure of the network.

Saadati et al. [29] presented the graph neural network (GNN) for generating a static cluster with an identical size. The developed GNN was utilised for balancing energy usage among the nodes that was mandatory in developing an energy effective WSN. Next is the distributed cluster head selection, which independently operated inside each cluster. The multi-hop routing was developed to avoid the long-distance communication which uses a certain level of energy. This work was mainly considered FND during CH selection, but appropriate fitness values were required to be considered to enhance the network lifecycle.

Hu et al. [30] developed the fuzzy logic and PSO-based clustering routing protocol, namely PFCRE, for enhancing energy efficiency and avoiding the energy hole issue. An energy effectiveness and load balance were considered in the PFCRE for generating the clustering based on an improved PSO. Next, an optimum route was identified by using a fuzzy inference system that was used for traffic load balancing and reducing energy usage. The distance among the nodes has to be considered in the clustering for additionally minimising the energy.

The gaps from the existing methods are ineffective cluster balancing, failure to achieve load balancing and incorporation of inappropriate objective values. In order to address these limitations, the M-PMARO based CH selection and routing is developed to achieve reliable data broadcasting in WSN.

3 | Preliminaries

This preliminaries section provides the information about the system model with its assumptions and energy expenditure model.

3.1 | System Model and Assumptions

Sensors are randomly positioned in the network area and the CHs are chosen from the sensors. Further, the clustering is done over the network whereas CHs from the cluster gathers the information from cluster members (CMs). The gathered data is aggregated by CH and broadcasted to BS via the optimum route. Figure 1 shows the architecture of clustering and routing in WSN.

The assumptions of this research are mentioned as follows:

- A sensors and BS in the WSN are static.
- Sensors have the capacity to report their position.
- All sensors are homogeneous, i.e., they have identical initial energy.
- Sensor's energy expenditure is influenced according to the interspace among the CH and BS.

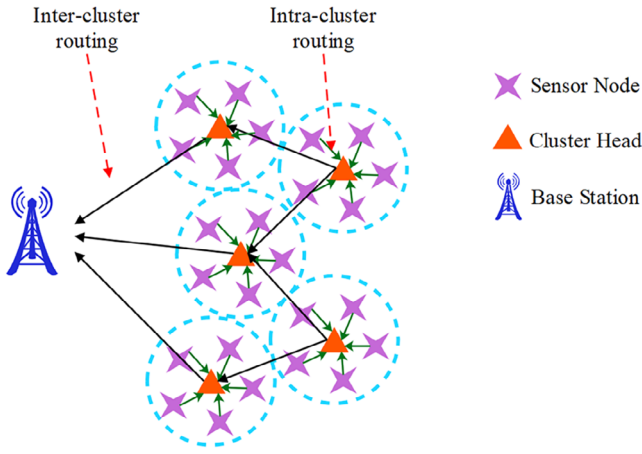


FIGURE 1 | Architecture of clustering and routing in WSN.

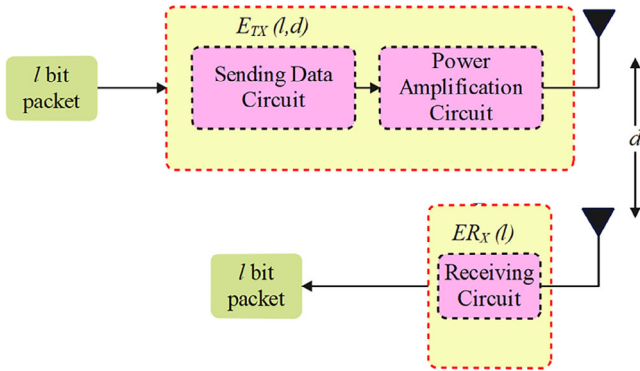


FIGURE 2 | Energy expenditure model.

- BS has higher computational power and it is not energy restricted in the network.

3.2 | Energy Expenditure Model

An energy expenditure of a sensor is divided into two different types, such as communication energy expenditure, which includes the energy utilised for sending and receiving the information, and data processing energy expenditure, which includes eliminating the unwanted data and performing data fusion. The communication environment, transmission distance and packet size influence the energy expenditure. The energy expenditure model is shown in Figure 2 according to the two perspectives, i.e., data transmission and data reception. If the distance between two sensors is d , Equations (1) and (2) are used to compute the energy expenditure of broadcasting and collecting the l bit data respectively [31].

$$E_{TX}(l, d) = \begin{cases} l \times E_{elec} + l \times \epsilon_{fs} \times d^2, & d \leq d_0 \\ l \times E_{elec} + l \times \epsilon_{mp} \times d^4, & d > d_0 \end{cases} \quad (1)$$

$$E_{RX}(l, d) = l \times E_{elec} \quad (2)$$

Where, energy expenditure for transmitting 1 bit data is denoted as E_{elec} ; ϵ_{fs} and ϵ_{mp} are the energy expenditure per bit for free space and multi-path fading models, respectively and the threshold value that separates the space model is denoted as d_0 . The value

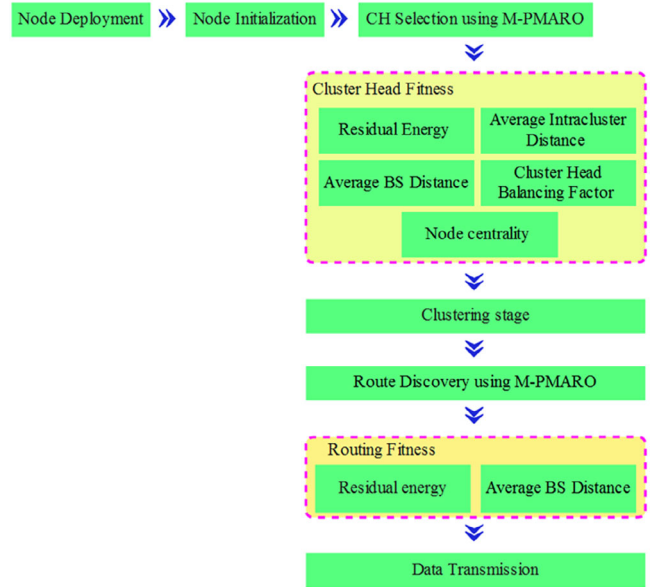


FIGURE 3 | Architecture of M-PMARO based CH and route discovery.

of d_0 is computed using Equation (3).

$$d_0 = \sqrt{\frac{\epsilon_{fs}}{\epsilon_{mp}}} \quad (3)$$

4 | Cluster Head Selection and Routing Using M-PMARO

This paper presented an effective CH and route discovery based on the M-PMARO algorithm for minimising the energy expenditure of the sensors. The developed research has three important phases, which are specified as follows: In the first phase, optimum CHs are chosen from the network based on M-PMARO and the clusters are created in the second phase. Finally, the optimum routes through the CHs are discovered by using the same M-PMARO. Figure 3 shows the architecture of M-PMARO based CH and route discovery.

4.1 | Network Initialisation Process

At first, the sensors are randomly dispersed in the network area along with the BS. The system model and its assumptions and their energy model considered in this research are specified in the previous sections. Further, the CHs are chosen and routing is done over the network using the M-PMARO.

4.2 | Process of CH Discovery Using M-PMARO

In this phase, optimum CHs among the sensors is chosen by using the M-PMARO. The typical ARO is inspired from the rabbit's existence schemes in the wild environment. The foraging and hiding schemes are done by ARO while the minimisation of energy was performed to perform the transformation between the strategies that were used for resolving the optimisation

problem. The rabbits of conventional ARO follow the other rabbit from the population in the exploration, which causes the invasive diversification pattern. Therefore, the experience based perturbed learning (EPL) approach is integrated for identifying the capable regions in the possible search space, which enhances the exploration. Moreover, the issue of local optima is avoided by incorporating the mutation strategy. The process of ARO is explained in the following sections:

4.2.1 | Initialisation for CHs

In this initialisation phase, the rabbit solutions are prepared with the random CMs from the network which are required to be chosen as CHs, where the total sensors are N . Consider, the rabbit i of M-PMARO is denoted as $z_i = (z_{i,1}, z_{i,2}, \dots, z_{i,D})$, where D is a dimension that equals the amount of CHs. The updates over the populations are specified in the following section.

4.2.2 | Detour Foraging (Exploration)

Consider, every rabbit in the ARO has its separate area with fewer burrows and grass. In the exploration process, a rabbit moves randomly away from its own zone for foraging food and observing what exists in the adjacent zone. This procedure is represented as detour foraging and it is shown using Equations (4) to (9).

$$z_i(t+1) = z_j(t) + A \times (z_i(t) - z_j(t)) + \text{round}(0.5 \times (0.05 \times R_1)) \times q_1, \quad (4)$$

$$A = L \times c \quad (5)$$

$$L = \left(e - e^{\left(\frac{t-1}{T}\right)^2} \right) \times \sin(2\pi R_2) \quad (6)$$

$$c(k) = \begin{cases} 1, & \text{if } k == g(l) \\ 0, & \text{otherwise} \end{cases}, \quad k = 1, \dots, D \text{ and } l = 1, \dots, [R_3 \times D] \quad (7)$$

$$g = \text{randperm}(D) \quad (8)$$

$$q_1 \sim P(0, 1) \quad (9)$$

where, $i, j = 1, \dots, P$ and $i \neq j$; $z_i(t+1)$ denotes the candidate location of i th rabbit in iteration $t+1$; rabbit i and j 's location of t th iteration are denoted as $z_i(t)$ and $z_j(t)$; the population size is P ; the current and maximum iteration are represented as t and T ; the dimension is D ; the ceiling operator is $\lceil \cdot \rceil$; rounding close to the number is $\text{round}(\cdot)$; an integer chosen among $[1, D]$ is $\text{randperm}(\cdot)$; the random values among $[0, 1]$ are R_1, R_2 & R_3 ; the standard normal distribution is q_1 and the step length is L .

4.2.3 | Conversion Between Exploration and Exploitation

At the beginning, the detour foraging is performed by the rabbits and the random hiding is recurrently performed in the subse-

quent phases of the searching operation. The rabbit energy (RE) denoted in Equation (10) is used to maintain the transformation between exploitation and exploration.

$$RE(t) = 4 \left(1 - \frac{t}{T} \right) \ln \frac{1}{R_4} \quad (10)$$

Where, a random value among the range of $[0, 1]$ is denoted as R_4 and RE is varied among the $[0, 2]$. If $RE > 1$, it means that the rabbit has higher energy for searching the zone of additional individuals, which is exploration. Otherwise, the rabbit accomplishes the exploitation, when $E \leq 1$ to escape from predation.

4.2.4 | Random Hiding (Exploitation)

In this exploitation, the rabbits are attacked or chased by predators. Consequently, various holes near the nest are developed for shelter. The capture probability is minimised by creating the D holes with the search space and selecting any hole randomly for hiding purposes. This random hiding is formulated in Equations (11) to (15).

$$z_i(t+1) = z_i(t) + A \times (R_5 \times b_{i,r}(t) - z_i(t)) \quad (11)$$

$$b_{i,r}(t) = z_i(t) + H \times g_r(k) \times z_i(t) \quad (12)$$

$$g_r(k) = \begin{cases} 1, & \text{if } k == \lceil R_6 \times D \rceil \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

$$H = \frac{T-t+1}{T} \times q_2 \quad (14)$$

$$q_2 \sim P(0, 1) \quad (15)$$

Where, a randomly elected hole of the i th rabbit within holes D used to hide at iteration t is $b_{i,r}(t)$; random value between $[0, 1]$ are specified as R_5 and R_6 , and standard normal distribution is q_2 .

4.2.5 | EPL Strategy

The EPL is initialised by computing the mean (z_{mean}) and deviation (z_{dev}) values of any solution selected randomly according to the optimum solution (z_{best}). Subsequently, the new solution is updated based on the mean and deviation with the perturbed term. Hence, the updated population is expressed as shown in Equations (16) to (20).

$$z_{\text{mean}}(t) = (z_{\text{best}} + z_1(t)) \quad (16)$$

$$z_{\text{dev}}(t) = \text{abs}(z_{\text{best}} - z_1(t)) \quad (17)$$

$$z_C(t) = z_{\text{mean}}(t) + \text{rand}_1 \times z_{\text{dev}}(t) \quad (18)$$

$$z_{\text{new}}(t) = z_C(t) + \text{rand}_2 \times (z_{\text{best}} - z_C(t)) + 0.95 \times (\text{rand}_3 - 0.5) \times \text{abs}(z_{\text{max},j} - z_{\text{min},j}) \quad (19)$$

$$z_{\text{max},j} = \max_j \{z_i(t)\}, \quad z_{\text{min},j} = \min_j \{z_i(t)\} \quad (20)$$

Where, the randomly selected arbitrary solution is represented as $z_1(t)$; random values based on the uniform distribution between $[0, 1]$ are $rand_1$, $rand_2$ and $rand_3$. Equation (19) accomplishes the perturbed solution between the dynamic limits $z_{\max,j}$ and $z_{\min,j}$. The incorporation of EPL is used to integrate the exploration and exploitation, which resulted in enhanced searching for optimum solution.

4.2.6 | Mutation Strategy

An unusual new individual is generated by using the mutation strategy and it leads to avoiding the issue of local optima. If the algorithm faces the local optima issue, the mutation operation is used to lead the solution out from it and enhance the exploration. The mutation operation is expressed in Equations (21) and (22).

$$F = 0.5 \times 2^{\frac{1-T}{T+1-T}} \quad (21)$$

$$z_i(t+1) = z_i(t) + F \times (z_a(t) - z_b(t)) \quad (22)$$

Where, mutation probability is denoted as F , and $z_a(t)$ and $z_b(t)$ are the randomly chosen population selected at iteration t . The iterative procedure of M-PMARO works until it reaches the termination criteria, i.e., maximum iterations fixed as 100.

4.3 | Multi-Objective CH Selection

The multiple objective values employed in CH discovery using M-PMARO are specified as follows.

4.3.1 | Residual Energy (f_1)

Network lifecycle depends on the energy usage, therefore it is important to minimise the energy expenditure. Hence, it is considered while choosing the CHs. This is computed according to the addition of a certain node's remaining energy. Since, an entire energy is required to be optimised, each objective is balance by opposite as shown in Equation (23).

$$f_1 = \frac{1}{\sum_{j=1}^D E_{CH_j}} \quad (23)$$

Where, E_{CH_j} denotes the energy exists in j^{th} CH.

4.3.2 | Average Intracluster Distance (f_2) and Average BS Distance (f_3)

An intracluster space is computed and expressed in Equation (24) as an addition of distances between the corresponding CH and the sensors. The energy expenditure of the network is decreased by intracluster distance. Moreover, the distance among the CH and BS to the entire sensors that exist in the CH is used to compute an average sink distance, which is expressed in Equation (25). Because, the space has a considerable effect on

energy expenditure, it is considered for decreasing the space to preserve the energy.

$$f_2 = \sum_{j=1}^D \left(\sum_{i=1}^{CM_j} \text{dis}(S_i, CH_j) / CM_j \right) \quad (24)$$

$$f_3 = \sum_{i=1}^D \text{dis}(CH_i, BS) \quad (25)$$

Where, distance among the sensors and CH is denoted as $\text{dis}(S_i, CH_j)$; distance among CH and BS is denoted as $\text{dis}(CH_i, BS)$ and CM_j is cluster member of the j^{th} cluster.

4.3.3 | Cluster Head Balancing Factor (f_4)

The clusters in the WSN are required to be balanced; there is a possibility that the network has small and huge clusters because of random sensor grouping in WSN. Accordingly, this balancing factor of Equation (26) is used to perform the balancing of energy expenditure.

$$f_4 = \sum_{i=1}^D \frac{AN}{m} - CM_i \quad (26)$$

Where, alive nodes denotes the AN .

4.3.4 | Node Centrality (f_5)

Node centrality expressed in Equation (27) is defined as how much a sensor is halfway located from the adjacent nodes. The nodes with better centrality are used to perform reliable communication over the network.

$$f_5 = \sum_{i=1}^D \frac{\sqrt{\frac{\sum_{j \in s} \text{dist}^2(i, j)}{s(i)}}}{\text{Network dimension}} \quad (27)$$

Where, amount of adjacent node in the i^{th} CH is s .

A weighted coefficient value is assigned to each objective value to transform the multiple objective values as single purpose value as shown in Equation (28).

$$\begin{aligned} \text{Clustering objectives} &= \mu_1 \times f_1 + \mu_2 \times f_2 + \mu_3 \times f_3 + \mu_4 \times f_4 \\ &+ \mu_5 \times f_5 \end{aligned}$$

Where, $\sum_{i=1}^5 \mu_i = 1$, $\mu_i \in (0, 1)$ (28)

Therefore, the nodes with higher energy with a lesser transmission distance, effective CH balancing and effective node centrality are chosen as CH over the network. The node with higher energy effectively performs data transmission and reception over the network as well as helps to avoid the node failure in the network. A node with a lesser distance helps to minimise energy expenditure, whereas the CHBF is used to balance the energy expenditure among the clusters. Moreover, the node with appropriate centrality helps to achieve reliable data broadcasting

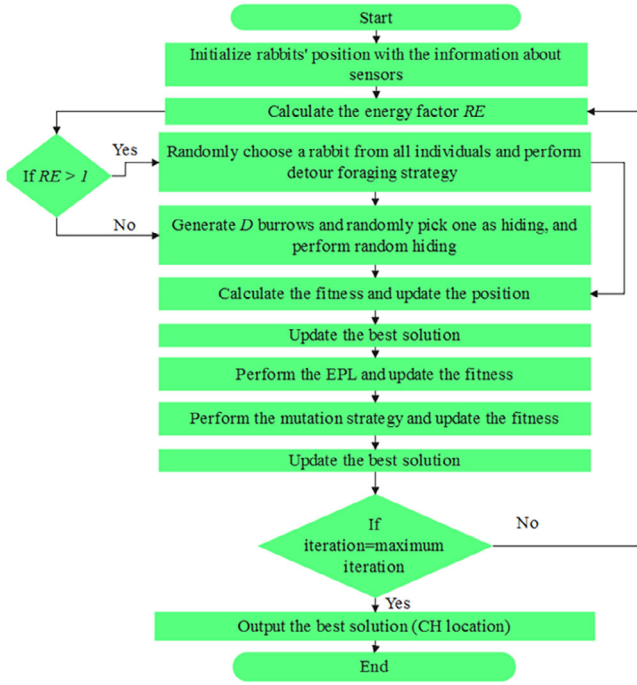


FIGURE 4 | CH discovery using M-PMARO.

in the network. The flowchart for CH discovery using M-PMARO is illustrated in Figure 4.

The pseudo code for CH discovery using M-PMARO is given the Algorithm 1.

4.4 | Clustering Stage

The BS broadcasts the ID of CH to an entire network and weight coefficients of every CH are computed once the CH discovery is completed by M-PMARO. The CMs join with the CH that has a higher weight coefficient. Accordingly, the clustering is done in the network. Equation (29) is used to calculate the weight values.

$$CH_{\text{weight}} = \frac{E_{\text{CH}}}{\text{dis}(S, CH) \times \text{dis}(CH, BS)} \quad (29)$$

Where, CH_{weight} represents the weight of the CH.

4.5 | Process of Route Discovery Phase Using M-PMARO

After forming the clusters over the network, the process of route identification is done using the M-PMARO. This routing phase uses single hop and multi hop path which help to reduce energy expenditure. If the transmitter CH is nearby with the BS, the information is straightly broadcasted to BS; else, the multi hop path is discovered via relay CHs to BS by using the M-PMARO. The process of multi hop routing using M-PMARO is specified as follows:

1. At first, the possible paths are used to initialise the solutions of M-PMARO, where the dimension is equal to relay CHs in the path.

ALGORITHM 1 | Pseudo code for CH discovery using M-PMARO.

1. Input: Initialise a set of rabbits randomly based on the node's information, population size (P), current iteration (t), maximum iteration ($T = 100$)
2. Compute each rabbit's fitness, i.e., objective function and determine the optimum solution
3. *for* $t = 1 : T$
4. *for* (each rabbit in P)
5. Compute the rabbit's energy using equation (10)
6. *if* $RE > 1$
7. From the overall population, select the rabbit randomly
8. Compute A based on equations (5) to (9).
9. Perform the exploration, i.e., detour foraging using equation (4).
10. Compute the rabbit's fitness.
11. Rabbit location is updated according to fitness.
12. *else*
13. Generate D amount burrows and choose any one randomly as hiding location based on equation (12).
14. Perform the random hiding based on equation (11).
15. Compute the rabbit's fitness.
16. Rabbit location is updated according to fitness.
17. *End if*
18. Update the optimum solution
19. Perform EPL using equations (16) to (20) and update the fitness
20. Perform mutation strategy using equations (21) and (22) and update the fitness
21. *End for*
22. $t = t + 1$
23. *End for*
24. Return the optimum solution, i.e., CH's location

2. The location update for path identification is identical to the iterative process in the previous section, whereas Equation (30) is the fitness function used in path identification.

$$\text{Routing objectives} = \delta_1 \times \sum_{i=1}^D E_{\text{CH}_i} + \delta_2 \times \sum_{i=1}^D \text{dis}(\text{CH}_i, \text{BS})$$

Where, $\sum_{i=1}^2 \delta_i = 1, \delta_i \in (0, 1)$ (30)

The formulated objective is employed for identifying the optimum shortest path via transmitter CHs to BS that helps to ensure the consistent data transfer. Further, the CH rotation is ensured in the WNS for additionally balancing the sensor's energy expenditure.

TABLE 1 | Parameter values.

Parameter	Value
Area	100 m × 100 m
Number of nodes	50, 100
Location of BS	(50, 100)
Initial energy	0.5J
ε_{mp}	0.0013pJ/bit/m ⁴
ε_{fs}	10pJ/bit/m ²
E_{elec}	50nJ/bit
Size of packet	4000 bits

5 | Simulations and Results

The M-PMARO is developed to ensure an optimum CH and path discovery in the WSN for achieving reliable communication. The experimental analysis is carried out to determine the efficiency of the proposed of M-PMARO.

5.1 | Experimental Setup and Evaluation Metrics

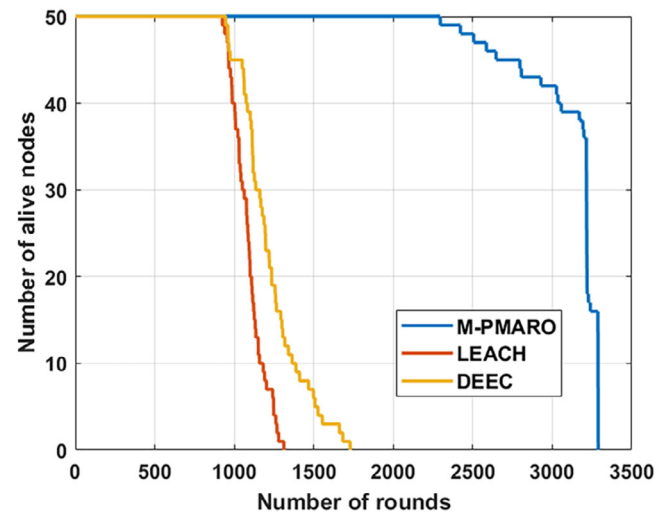
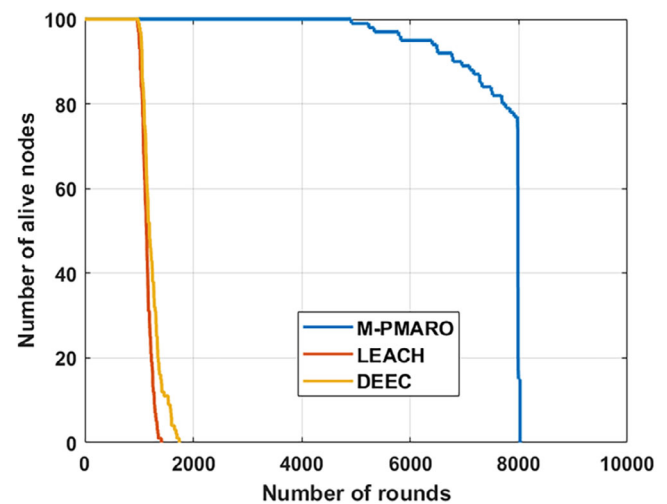
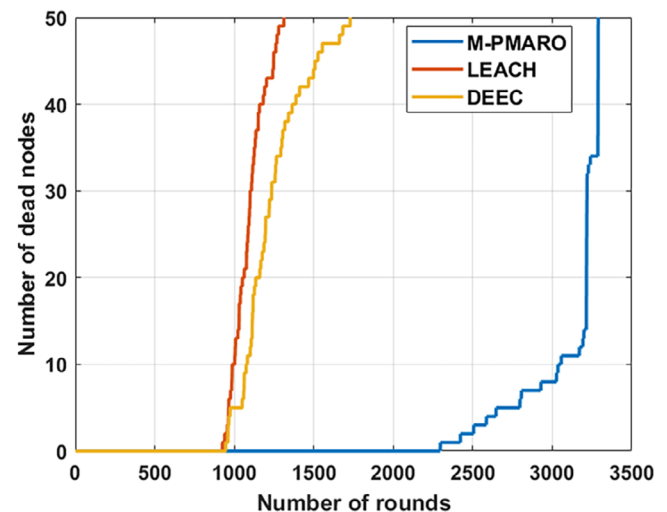
For simulating the M-PMARO in MATLABR 2020b software, the analysis is performed on an i7 processor, Windows 10 OS and 16GB RAM. The parameter values of the simulation for M-PMARO are shown in Table 1. The M-PMARO is analysed using an alive node count, dead nodes, energy expenditure, throughput, data received in BS and network lifecycle.

5.2 | Evaluation of Results With Classical Approaches

Initially, the classical methods such as DEEC and LEACH are used to compare the M-PMARO. These classical methods of DEEC and LEACH are designed for the same parameter values as Table 1.

5.2.1 | Alive Nodes and Dead Nodes Analysis

Alive nodes are defined as the sensors that have energy to perform the data transmission and reception over the network. Moreover, the dead nodes are generally opposite of the alive nodes, i.e., they denote the sensors that do not have energy to perform the communication. The evaluation of alive nodes for the M-PMARO with 50 and 100 nodes is shown in Figures 5 and 6. Similarly, the evaluation of dead nodes for the M-PMARO with 50 and 100 nodes is shown in Figures 7 and 8. These figures represent that the M-PMARO achieves higher number of alive nodes than the classical methods. The proposed M-PMARO with higher alive nodes is obtained by decreasing the energy. The optimum CH discovery of M-PMARO with a lesser broadcasting distance and effective centrality and cluster balancing leads to reduced energy. In M-PMARO, the chosen CH with effective centrality helps to perform the load balancing, which helps to attain uniform

**FIGURE 5** | Alive nodes per round for 50 nodes.**FIGURE 6** | Alive nodes per round for 100 nodes.**FIGURE 7** | Dead nodes per round for 50 nodes.

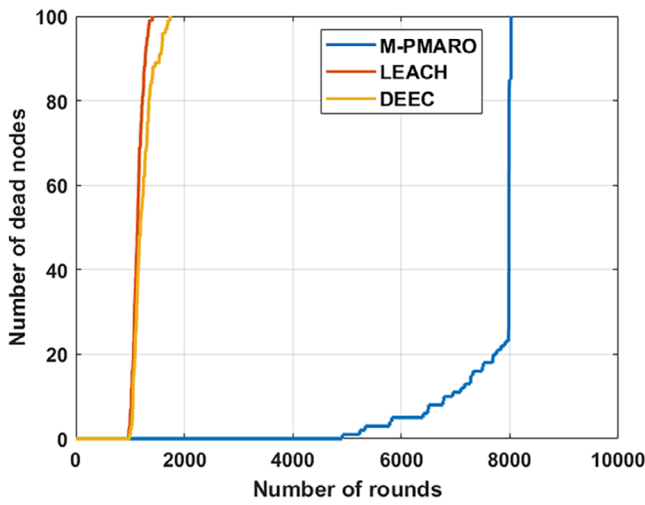


FIGURE 8 | Dead nodes per round for 100 nodes.

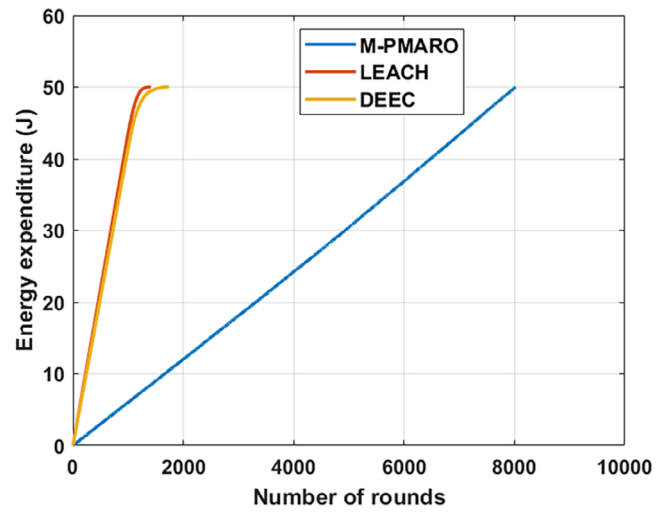


FIGURE 10 | Energy expenditure per round for 100 nodes.

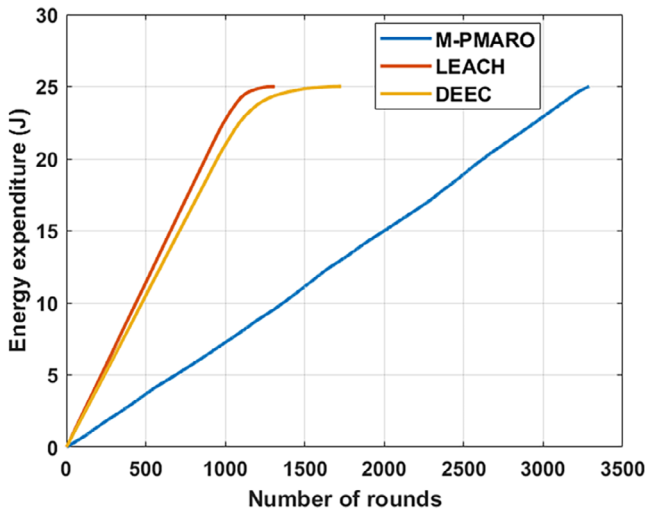


FIGURE 9 | Energy expenditure per round for 50 nodes.

distribution of energy. Moreover, the energy balancing among the clusters is done by considering the CHBF during CH discovery.

5.2.2 | Energy Expenditure Analysis

It is the amount of energy utilised during the data transmission and reception in the WSN. The resultant graphs for energy expenditure of M-PMARO for 50 and 100 nodes are shown in Figures 9 and 10. As shown in Figures 9 and 10, it is observed that the energy expenditure of the M-PMARO is less than the classical methods. The M-PMARO achieves less energy expenditure by ensuring an optimum CH and route discovery. The CHBF considered by M-PMARO is used to balance the energy among the clusters. The minimisation of broadcasting distance and an effective load balancing achieved by the node centrality factor using M-PMARO leads to minimise the energy usage. Further, the combination of single and multi hop path discovery using M-PMARO additionally helps to minimise the energy expenditure.

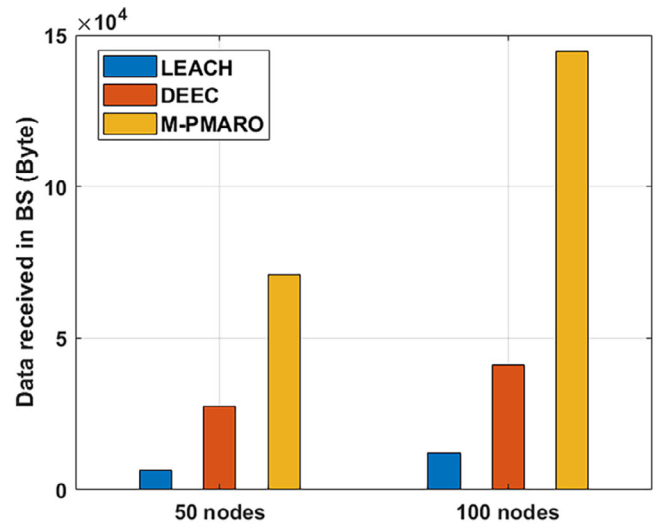


FIGURE 11 | Data received in BS analysis.

5.2.3 | Data Received in BS and Throughput Analysis

The key procedure is to perform effective data broadcasting over the network. Therefore, throughput and data received in BS are supportive for identifying the efficacy of the M-PMARO. Here, the total data packets effectively collected in the BS are denoted as throughput. The resultant graphs for data received in BS and throughput of M-PMARO for 50 and 100 nodes are shown in Figures 11 and 12 respectively. As shown in Figures 11 and 12, it is observed that the M-PMARO has improved data delivery compared to the classical methods. The data delivery of the M-PMARO is improved by avoiding the node failure and reducing the energy. The information loss over the network is minimised by considering the energy factor while performing the CH and path discovery in M-PMARO. The reliable data broadcasting is obtained by considering the centrality factor at CH selection and by performing an efficient multi hop routing in the network.

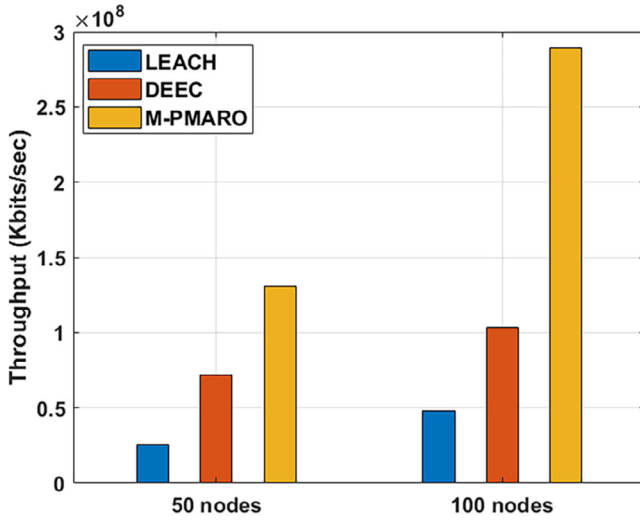


FIGURE 12 | Throughput analysis.

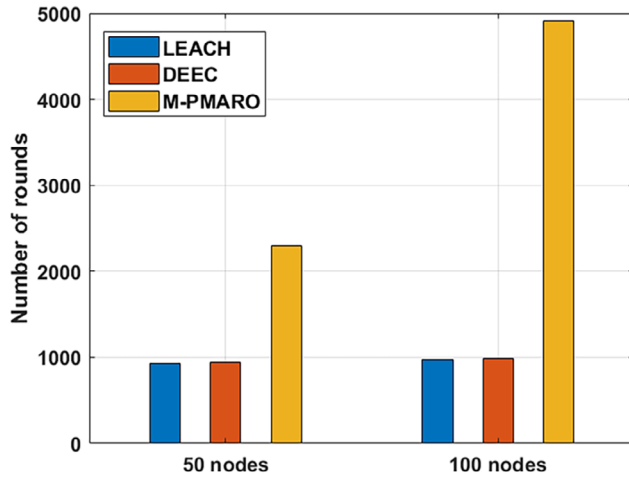


FIGURE 13 | FNE analysis.

5.2.4 | Network Lifecycle

Network lifecycle is the interval between the node initialisation at WSN and all the nodes in the network expires, whereas it is analysed in three different factors: (a) first node expiration (FNE), (b) half node expiration (HNE) and (c) last node expiration (LNE). FNE is the round at which the first node expires, HNE is the round at which half of the node expires and LNE is the round at which all node expires in the network. The resultant graphs of M-PMARO's network lifecycle for FNE, HNE and LNE are shown in Figures 13, 14 and 15 respectively. As shown in Figures 13, 14 and 15, it is observed that the M-PMARO has a higher network lifecycle than the classical methods. The higher network lifecycle of M-PMARO is achieved by reducing the energy. The energy expenditure of the M-PMARO is minimised according to the below schemes: the lesser broadcasting distance and optimum shortest path are used to preserve the energy of the nodes. Moreover, the balancing among the clusters and an effective load balancing of the CHs further minimises the energy expenditure.

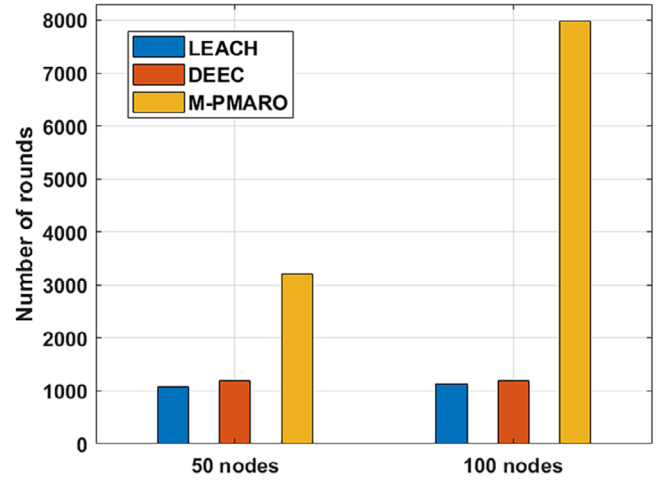


FIGURE 14 | HNE analysis.

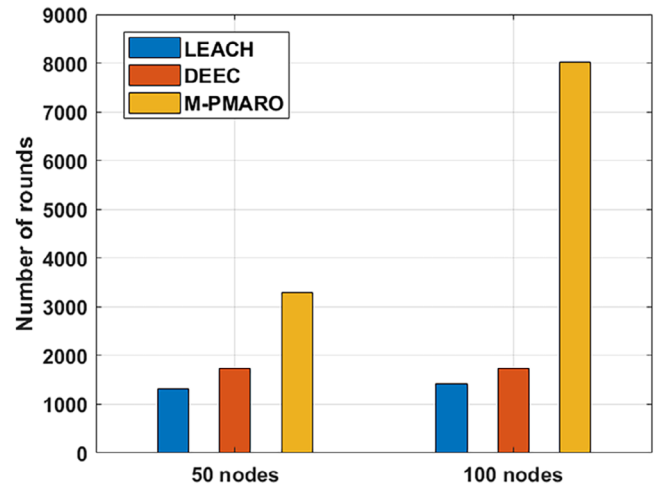


FIGURE 15 | LNE analysis.

5.3 | Evaluation of Results With Optimisation Approaches

This section offers the evaluation of M-PMARO with some other optimisation approaches such as coati optimisation algorithm (COA) and pelican optimisation algorithm (POA). The network lifecycle is one of the important parameters during the design of energy efficient WSN, therefore an evaluation of FNE, HNE and LNE is provided in the Figures 16, 17 and 18 respectively. This evaluation demonstrates that the proposed M-PMARO has better performance than the COA and POA because of the optimum solutions obtained via the searching procedure. The integration of EPL is used to detect the capable regions in the possible search space that lead to enhanced exploration while the mutation strategy is used to avoid the issue of local optima.

5.4 | Evaluation of Results With Existing Approaches

The existing approaches such as FGF [23], EBPSO [24], IBOA [25], GNN [29] and PFCRE [30] are used to know the viability of the

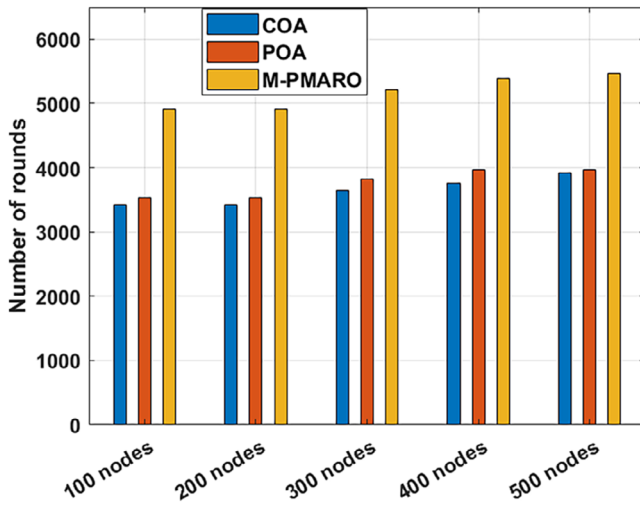


FIGURE 16 | FNE analysis for optimisation approaches.

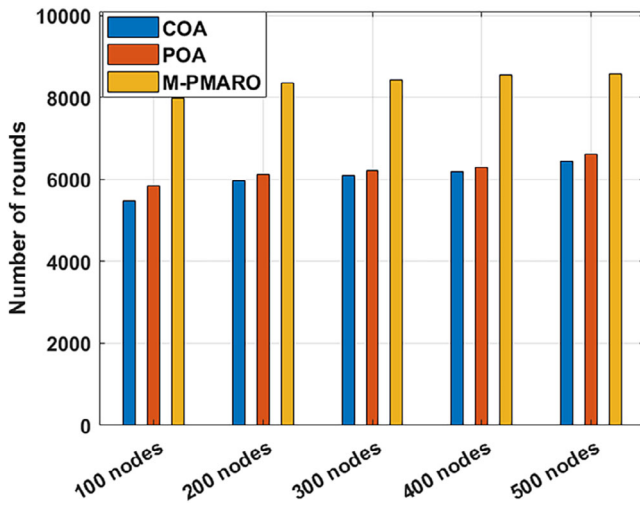


FIGURE 17 | HNE analysis for optimisation approaches.

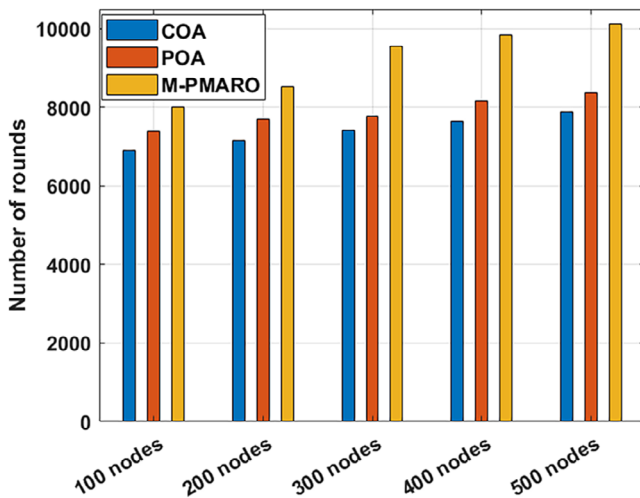


FIGURE 18 | LNE analysis for optimisation approaches.

M-PMARO method. The comparison of M-PMARO is done for different cases which are specified in the Table 2.

The comparative evaluation of M-PMARO is shown in Tables 3–7. In that, Table 3 shows the comparison with FGF [23] for Case 1, Table 4 shows the comparison with EBPSO [24] for Case 2, Table 5 shows the comparison with IBOA [25] for Cases 3 & 4, Table 6 shows the comparison with GNN [29] for Case 5, and Table 7 shows the comparison with PFCRE [30] for Case 6. As shown in the tables, it is observed that the proposed M-PMARO achieves a higher number of alive nodes and data packets received in BS. For example, an alive node of the M-PMARO for case 1 at round 2000 is 100, whereas the FGF [23] has lesser number of alive nodes, i.e., 28. The alive nodes of the M-PMARO are increased by preserving the energy expenditure of nodes according to the energy balancing among the clusters and an effective load balancing between the clusters. Because, an effective load balancing among the nodes benefits in uniform energy dissemination over the network. Further, an optimum shortest path from M-PMARO leads to lessen the energy of the nodes. On the other hand, an appropriate CH and path from the M-PMARO are used to ensure reliable data communication. Moreover, the node failure is avoided by M-PMARO which additionally helps to enhance the communication.

6 | Discussion

This section offers the discussion about the M-PMARO developed for reliable communication over the WSN. Initially, the developed M-PMARO is evaluated with the classical methods of DEEC and LEACH. Compared to the classical methods, the M-PMARO achieves better performance due to its effective cluster balancing, centrality and multi-hop routing. However, the classical methods of DEEC and LEACH obtain lesser performance, because they considered only the energy parameter during the CH selection. For effective energy balancing, the distance, cluster balancing and node centrality are required to be considered in WSN. Moreover, these classical methods perform only single hop data broadcasting, because of that, it exhaust their energy rapidly and obtain a lesser network lifecycle. Next, the enhanced exploration and mitigation of the local optima issue using EPL and the mutation strategy in M-PMARO made it is effective than the COA and POA. Additionally, the proposed M-PMARO achieved better performance than the FGF, EBPSO, IBOA, GNN and PFCRE. The load and energy balancing among the clusters along with the optimum multi hop path with a lesser distance are used to preserve the energy expenditure of the nodes. Further, the mitigation of node failure in M-PMARO is used to enhance the data delivery of the WSN. However, the security is considered as a challenging factor in the WSN, due to the deployment of wireless sensors in an open medium.

7 | Conclusion

This research has improved the multiobjective CH and routing path selection in WSN. An important objective of this research is to improve the network lifecycle and data delivery over the WSN. This research proposes the M-PMARO for choosing the optimum CH that leads to balancing the energy and achieving reliable

TABLE 2 | Information about cases.

Parameters	Cases					
	1	2	3	4	5	6
Nodes	100	100	400	400	500	200
Area	100 m × 100 m	100 m × 100 m	250 m × 250 m	250 m × 250 m	300 m × 300 m	800 m × 800 m
BS location	(50, 50)	(50, 100)	(100, 100)	(200, 200)	(150, 500)	(400, 400)

TABLE 3 | Comparison for case 1.

Rounds	Alive nodes	
	FGF [23]	M-PMARO
500	100	100
1000	53	100
1500	34	100
2000	28	100

TABLE 4 | Comparison for case 2.

Rounds	Alive nodes	
	EBPSO [24]	M-PMARO
200	100	100
400	100	100
600	100	100
800	100	100
1000	100	100
1200	46	100
1400	0	100

TABLE 5 | Comparison for cases 3 and 4.

Cases	Methods	Data received in BS (Byte)
Case 3	IBOA [25]	1.9×10^6
	M-PMARO	2.83×10^7
Case 4	IBOA [25]	1.87×10^6
	M-PMARO	2.39×10^7

data transfer. The clustering of normal nodes to the specific CHs is done by considering the weight coefficient, which includes the remaining energy and distance. Further, the incorporation of single and multi-hop paths is accomplished for reducing the energy of the nodes. Specifically, the multi-hop route selection is done by using the M-PMARO for enhancing the network lifecycle and data broadcasting in the network. The simulations and results represented that the M-PMARO outperforms the FGF, EBPSO, IBOA, GNN and PFCRE. For instance, the alive node of M-PMARO for 1200 rounds is 100, which is higher than the EBPSO. In future, the security of the data transmission can be obtained by developing trust based routing in WSN.

TABLE 6 | Comparison for case 5.

Rounds	Alive nodes	
	GNN [29]	M-PMARO
100	500	500
200	500	500
300	400	500
400	0	500

TABLE 7 | Comparison for case 6.

Rounds	Alive nodes	
	PFCRE [30]	M-PMARO
100	185	200
200	120	200
300	38	200
400	23	200
500	12	172
600	0	168

Author Contributions

Babiyola Arulanandam: methodology, software, review and editing of original draft. **Parameshchari B. D.:** resources, write-up of original draft, formal analysis, investigation. **Rocio Pérez de Prado:** conceptualisation, validation. **Khalid Nazim Abdul Sattar:** project administration, data curation, resources.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

References

1. N. Ma, H. Zhang, H. Hu, and Y. Qin, "ESCVAD: An Energy-Saving Routing Protocol Based on Voronoi Adaptive Clustering for Wireless Sensor Networks," *IEEE Internet of Things Journal* 9, no. 11 (2021): 9071–9085, <https://doi.org/10.1109/JIOT.2021.3120744>.
2. S. Jagadeesh and I. Muthulakshmi, "Dynamic Clustering and Routing Using Multi-Objective Particle Swarm Optimization With Levy Distribution for Wireless Sensor Networks," *International Journal of*

- Communication Systems* 34, no. 13 (2021): e4902, <https://doi.org/10.1002/dac.4902>.
3. E. Heidari, A. Movaghar, H. Motameni, and B. Barzegar, "A Novel Approach for Clustering and Routing in WSN Using Genetic Algorithm and Equilibrium Optimizer," *International Journal of Communication Systems* 35, no. 10 (2022): e5148, <https://doi.org/10.1002/dac.5148>.
 4. A. Muniraj and J. E. Jeyaswamidoss, "An Enhanced Gray Wolf Optimization for Cluster Head Selection in Wireless Sensor Networks," *International Journal of Communication Systems* 35, no. 15 (2022): e5300, <https://doi.org/10.1002/dac.5300>.
 5. G. Santhosh and K. V. Prasad, "Energy Optimization Routing for Hierarchical Cluster Based WSN Using Artificial Bee Colony," *Measurement: Sensors* 29 (2023): 100848, <https://doi.org/10.1016/j.measen.2023.100848>.
 6. V. Narayan, A. K. Daniel, and P. Chaturvedi, "E-FEERP: Enhanced Fuzzy Based Energy Efficient Routing Protocol for Wireless Sensor Network," *Wireless Personal Communications* 131 (2023): 371–398, <https://doi.org/10.1007/s11277-023-10434-z>.
 7. D. L. Balasubramaniana and V. Govindasamy, "Design of Improved Deer Hunting Optimization Enabled Multihop Routing Protocol for Wireless Sensor Networks," *International Journal of Cognitive Computing in Engineering* 4 (2023): 363–372, <https://doi.org/10.1016/j.ijcce.2023.10.002>.
 8. L. Abbad, A. Nacer, H. Abbad, M. T. Brahim, and N. Zioui, "A Weighted Markov-Clustering Routing Protocol for Optimizing Energy Use in Wireless Sensor Networks," *Egyptian Informatics Journal* 23, no. 3 (2022): 483–497, <https://doi.org/10.1016/j.eij.2022.05.001>.
 9. A. Ojha and P. Chanak, "Multiobjective Gray-Wolf-Optimization-Based Data Routing Scheme for Wireless Sensor Networks," *IEEE Internet of Things Journal* 9, no. 6 (2021): 4615–4623, <https://doi.org/10.1109/JIOT.2021.3105425>.
 10. S. Ramalingam, S. Dhanasekaran, S. S. Sinnasamy, A. O. Salau, and M. Alagarsamy, "Performance Enhancement of Efficient Clustering and Routing Protocol for Wireless Sensor Networks Using Improved Elephant Herd Optimization Algorithm," *Wireless Networks* 30, no. 3 (2024): 1773–1789.
 11. X. Guo, Y. Ye, L. Li, R. Wu, and X. Sun, "WSN Clustering Routing Algorithm Combining Sine Cosine Algorithm and Lévy Mutation," *IEEE Access* 11 (2023): 22654–22663, <https://doi.org/10.1109/ACCESS.2023.3252027>.
 12. S. Nasirian, P. Pierleoni, A. Belli, M. Mercuri, and L. Palma, "Pizzza: A Joint Sector Shape and Minimum Spanning Tree-Based Clustering Scheme for Energy Efficient Routing in Wireless Sensor Networks," *IEEE Access* 11 (2023): 68200–68215, <https://doi.org/10.1109/ACCESS.2023.3291915>.
 13. M. Mozaffari, S. M. Mazinani, and A. A. Khazaei, "An Energy Efficient Grid-Based Clustering Algorithm Using Type-3 Fuzzy System in Wireless Sensor Networks," *Wireless Networks* 31 (2024): 109–125.
 14. N. Maliseti and V. K. Pamula, "Energy Aware Cluster-Based Routing in WSN Using Hybrid pelican-Blue Monkey Optimization Algorithm," *Evolutionary Intelligence* 17 (2024): 2555–2575.
 15. I. K. Shah, T. Maity, Y. S. Dohare, D. Tyagi, D. Rathore, and D. S. Yadav, "ICIC: A Dual Mode Intra-Cluster and Inter-Cluster Energy Minimization Approach for Multihop WSN," *IEEE Access* 10 (2022): 70581–70594, <https://doi.org/10.1109/ACCESS.2022.3188684>.
 16. G. S. Arunachalam, G. Ramalingam, and R. Nanjappan, "A Classy Energy Efficient Spider Monkey Optimization Based Clustering and Data Aggregation Models for Wireless Sensor Network," *Concurrency and Computation: Practice and Experience* 35, no. 2 (2023): e7492, <https://doi.org/10.1002/cpe.7492>.
 17. Y. Gong, X. Guo, and G. Lai, "A Centralized Energy-Efficient Clustering Protocol for Wireless Sensor Networks," *IEEE Sensors Journal* 23, no. 2 (2022): 1623–1634, <https://doi.org/10.1109/JSEN.2022.3224180>.
 18. S. J. Pratha, V. Asanambigai, and S. R. Mugunthan, "A Modified Whale-Dragonfly Algorithm and Self-Adaptive Cuckoo Search-Based Clustering Strategy for Augmenting Network Lifetime in Wireless Sensor Networks," *International Journal of Communication Systems* 36, no. 9 (2023): e5482, <https://doi.org/10.1002/dac.5482>.
 19. M. Elshrkawey, H. Al-Mahdi, and W. Atwa, "An Enhanced Routing Algorithm Based on a Re-Position Particle Swarm Optimization (RA-RPSO) for Wireless Sensor Network," *Journal of King Saud University—Computer and Information Sciences* 34, no. 10 (2022): 10304–10318, <https://doi.org/10.1016/j.jksuci.2022.10.022>.
 20. S. K. Barnwal, A. Prakash, and D. K. Yadav, "Improved African Buffalo Optimization-Based Energy Efficient Clustering Wireless Sensor Networks Using Metaheuristic Routing Technique," *Wireless Personal Communications* 130, no. 3 (2023): 1575–1596, <https://doi.org/10.1007/s11277-023-10345-z>.
 21. S. Chaurasia, K. Kumar, and N. Kumar, "EEM-CRP: Energy Efficient Meta-Heuristic Cluster Based Routing Protocol for WSNs," *IEEE Sensors Journal* 23, no. 23 (2023): 29679–29693, <https://doi.org/10.1109/JSEN.2023.3322631>.
 22. R. Mishra and R. K. Yadav, "Energy Efficient Cluster-Based Routing Protocol for WSN Using Nature Inspired Algorithm," *Wireless Personal Communications* 130, no. 4 (2023): 2407–2440, <https://doi.org/10.1007/s11277-023-10385-5>.
 23. K. N. Dattatraya and K. R. Rao, "Hybrid Based Cluster Head Selection for Maximizing Network Lifetime and Energy Efficiency in WSN," *Journal of King Saud University—Computer and Information Sciences* 34, no. 3 (2022): 716–726, <https://doi.org/10.1016/j.jksuci.2019.04.003>.
 24. Y. Xiuwu, Z. Zixiang, P. Wei, and L. Yong, "A Novel Multi-Hop Clustering Routing Algorithm Based on Particle Swarm Optimization for Wireless Sensors Networks," *Wireless Personal Communications* 130, no. 2 (2023): 935–956, <https://doi.org/10.1007/s11277-023-10314-6>.
 25. S. Janakiraman, "Improved Bat Optimization Algorithm and Enhanced Artificial Bee Colony-Based Cluster Routing Scheme for Extending Network Lifetime in Wireless Sensor Networks," *International Journal of Communication Systems* 36, no. 5 (2023): e5428, <https://doi.org/10.1002/dac.5428>.
 26. S. M. M. H. Daneshvar and S. M. Mazinani, "On the Best Fitness Function for the WSN Lifetime Maximization: A Solution Based on a Modified Salp Swarm Algorithm for Centralized Clustering and Routing," *IEEE Transactions on Network and Service Management* 20, no. 4 (2023): 4244–4254.
 27. S. Ambareesh, H. C. Kantharaju, and M. Sakthivel, "A Novel Fuzzy TOPSIS Based Hybrid Jarratt Butterfly Optimization for Optimal Routing and Cluster Head Selection in WSN," *Peer-to-Peer Networking Appl* 16, no. 5 (2023): 2512–2524, <https://doi.org/10.1007/s12083-023-01517-6>.
 28. P. Rawat, P. Kumar, and S. Chauhan, "Fuzzy Logic and Particle Swarm Optimization-Based Clustering Protocol in Wireless Sensor Network," *Soft Computing* 27, no. 9 (2023): 5177–5193, <https://doi.org/10.1007/s00500-023-07833-6>.
 29. M. Saadati, S. M. Mazinani, A. A. Khazaei, and S. J. S. M. Chabok, "Energy Efficient Clustering for Dense Wireless Sensor Network by Applying Graph Neural Networks With Coverage Metrics," *Ad Hoc Networks* 156 (2024): 103432.
 30. H. Hu, X. Fan, C. Wang, T. Wang, and Y. Deng, "Particle Swarm Optimization and Fuzzy Logic Based Clustering and Routing Protocol to Enhance Lifetime for Wireless Sensor Networks," *Cluster Computing* 27 (2024): 9715–9734.
 31. D. Wang, Q. Wu, and M. Hu, "Hierarchical Energy-Saving Routing Algorithm Using Fuzzy Logic in Wireless Sensor Networks," *EURASIP Journal on Information Security* 2023, no. 1 (2023): 9, <https://doi.org/10.1186/s13635-023-00144-1>.